

OPAL – Omnidirectional Path-efficient Aerial 3D expLoration

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Abstract—Autonomous exploration must balance distance efficiency against the cost of acquiring and processing information at competing frontiers. We present Omnidirectional Path-efficient Aerial 3D expLoration (OPAL), an aerial exploration framework that performs an in-place 360° whole-vehicle yaw scan at ambiguous branch points before committing to a frontier. Across six Habitat-Matterport3D (HM3D) scenes, OPAL-NFP₁₀ improves coverage–distance area under the curve (AUC) by 10.9% and reduces travel distance by 7.0% relative to EDEN, but requires 3.7× EDEN’s elapsed time. Representative onboard trials in two indoor layouts demonstrate feasibility and show travel reductions of up to 25.5% relative to FALCON, at the cost of a 68.8% increase in elapsed time. Overall, OPAL trades time spent on whole-vehicle rotation for more distance-efficient exploration.

I. INTRODUCTION

Autonomous exploration is essential for robots mapping previously unseen indoor environments. Frontier methods provide a simple and robust interface between known and unknown space [1], but selecting among feasible frontiers remains difficult. A nearby frontier can lead to a dead end, whereas a slightly farther one can expose a substantially larger region.

Global aerial exploration includes receding-horizon and sampling-based methods [2], [3], [4], incremental frontier tours such as FUEL [5], mixed frontier sampling in FSMP [6], hierarchical planning [7], [8], and dynamic-environment planning [9]. Recent systems such as FALCON [10] and EDEN [11] use increasingly rich global guidance, while learned active-mapping policies include GLEAM and GenNBV [12], [13]. Wide-field sensing offers a complementary approach, but the sensing policy matters. Lantern-Explorer/FUEL-360 exploits native wide-field-of-view (FoV) LiDAR [14], VRExplorer co-optimizes the direction of a steered camera [15], and active-LiDAR FLARE uses a dedicated orientation actuator [16]. In this paper, we present Omnidirectional Path-efficient Aerial 3D expLoration (OPAL), an aerial exploration framework that performs an in-place 360° whole-vehicle yaw scan at ambiguous branch points before committing to a frontier. Specifically, we examine whether a whole-vehicle pan can reduce subsequent travel distance enough to justify the resulting increase in elapsed time.

OPAL retains FALCON’s geometric mapping, planning, and control stack but inserts a decision layer between frontier generation and target commitment. At a branch point, the vehicle pauses for an in-place 360° yaw pan and selects a target from a reachable-vicinity set. This interface enables a comparison between a geometric rule and selectors based on a large language model (LLM) or vision-language

model (VLM), inspired by high-level visual frontier guidance in terrestrial exploration [17]. Semantic exploration and embodied-model systems span object-labeled planning and general multimodal control [18], [19]. Newer task-driven systems address adaptive object navigation, onboard aerial language navigation, and open-world structure scanning [20], [21], [22]. In comparison, our study is limited to local frontier arbitration from geometric and visual evidence. It does not evaluate broader semantic priors, task reasoning, or long-horizon memory.

This work contributes (1) a branch-triggered panoramic sensing and frontier-selection layer for 3D aerial exploration and (2) simulation and hardware evidence that characterizes the distance–time trade-off of in-place panoramic sensing. The event-triggered sensing policy and system characterization constitute the central contribution. LLM- and VLM-based selectors serve as secondary variants.

II. METHOD

OPAL retains FALCON’s sensing, MapServer, FrontierFinder, target validation, trajectory generation, and execution components. It adds a reachable-vicinity filter, a decision gate, a local selector, and lightweight branch memory for recovery (Fig. 1). Pose and depth are fused into the voxel map, while the inherited truncated signed distance field (TSDF) and Euclidean signed distance field (ESDF) representations support collision checking and path planning. RGB images provide visual evidence during a pan.

Let F_t be the active frontier viewpoints, $x_t \in \mathbb{R}^3$ the robot position, and $a_f \in \mathbb{R}^3$ the average position of frontier f . For vicinity radius r_v , OPAL forms the reachable-vicinity set $V_{r_v,t}$, i.e., the set of reachable frontier viewpoints within radius r_v of the robot at time t :

$$V_{r_v,t} = \{f \in F_t \mid \|a_f - x_t\| \leq r_v, q_t(f) = 1\}, \quad (1)$$

where $q_t(f)$ indicates that f has sufficient clearance and a feasible planner path. Small r_v values force nearby exploration. Increasing r_v relaxes that constraint, and $r_v \rightarrow \infty$ permits global choice.

The cardinality of $V_{r_v,t}$ controls the decision gate. If the set is empty, the recovery procedure revisits an unresolved earlier branch. With one candidate, the robot navigates to it directly. With multiple candidates, the robot hovers, performs a 360° yaw pan, and invokes the selector, which can be either geometric or model-based. During the continuous yaw, RGB frames are sampled from the live stream rather than stitched into a panorama. Each frontier is associated with the frame whose viewing direction has the smallest angular difference from the frontier bearing. The match is discarded if that

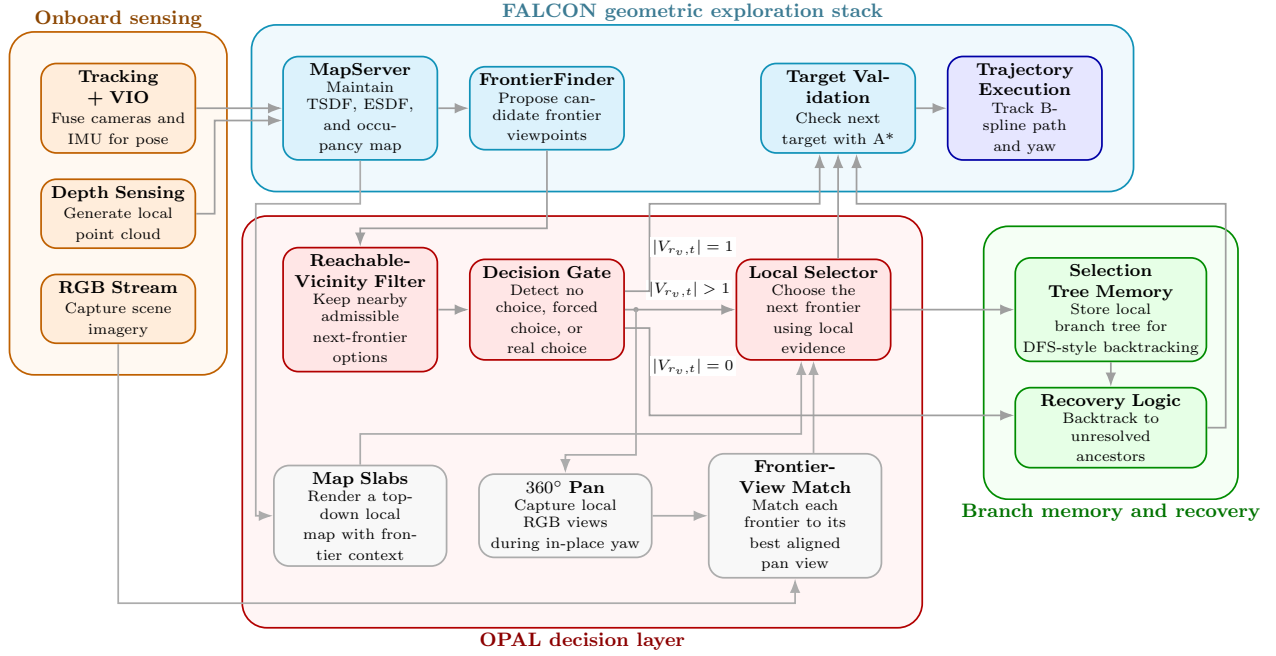


Fig. 1. Block diagram of OPAL, which adds local frontier selection to the FALCON [10] exploration stack. Mapping, frontier generation, target validation, planning, and trajectory execution are retained from FALCON [10], while a 360° pan is performed only when multiple reachable candidates compete.

difference exceeds 45°. A top-down map slab shows free and occupied space, the recent trajectory, the current pose, decision states, and candidate frontiers (Fig. 2).

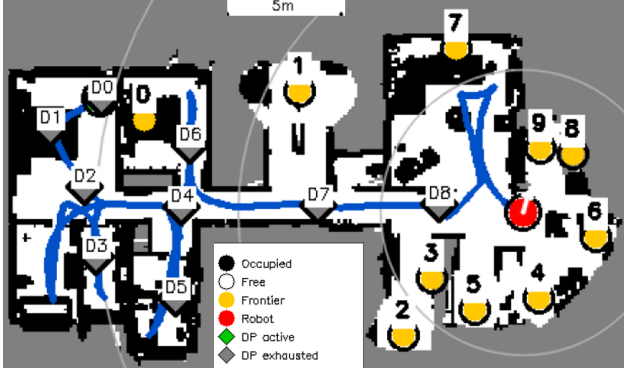


Fig. 2. A local map slab used at a decision point. Free and occupied cells are white and black, respectively; the recent trajectory is blue, the robot pose is red, candidate frontiers are yellow, the active decision point is green, and exhausted decisions are gray.

After triggering a pan, the robot acquires depth and RGB data during the yaw. The post-pan pipeline can then fuse the new depth observations, regenerate the frontier set before target validation, compute A* cost, select a target, and resume execution.

The primary policy, OPAL-NFP (Nearest Frontier + Pan), selects the candidate with the lowest A* planner cost after the pan and uses no RGB or semantic evidence. Its no-pan variant, OPAL-NF, applies the same rule without the yaw observation. The model-based OPAL-V supplies the map slab, matched frontier images, and planner costs to a VLM, while OPAL-OF omits the slab, and OPAL-L supplies only

textual metadata to an LLM. The latter three OPAL variants are all model-based and use Gemini 2.0 Flash. The policy remains fixed within each run. Selected frontiers are stored with their unresolved alternatives, allowing the recovery procedure to backtrack when local progress is exhausted, in the spirit of depth-first search and frontier-tree exploration [23], [24].

III. EXPERIMENTS

We evaluate six HM3D scenes [25], namely 130, 195, 231, 417, 666, and 804. Their textured meshes allow the visual variants to collect frontier-aligned images. Before simulation, we repair wall holes and internal cavities in Blender [26] to make the environments traversable by an aerial robot. These modifications may alter scene topology. Each method uses the same 5 meter depth range and 50 matched trials per map. Each of 25 validated free-space starts is repeated twice. A run terminates when no frontier remains available. We compare against FALCON [10] and EDEN [11]. Other omnidirectional systems, including Lantern/FUEL-360 [14], are relevant, but we could not obtain executable implementations suitable for the shared benchmark and finite-FoV sensor configuration. FUEL-360 [14] and learned next-best-view methods such as GLEAM [12] and GenNBV [13] remain priorities for future matched evaluation.

Let $C_m(d)$ denote the coverage achieved by method m after traveling distance d . We report terminal travel distance and $AUC_m = \int_0^D C_m(d) dd$, where D is the largest terminal distance among the compared methods. For methods that terminate before D , coverage is held at its terminal value. AUC has units of percentage-point meters and measures how quickly coverage accumulates with distance traveled.

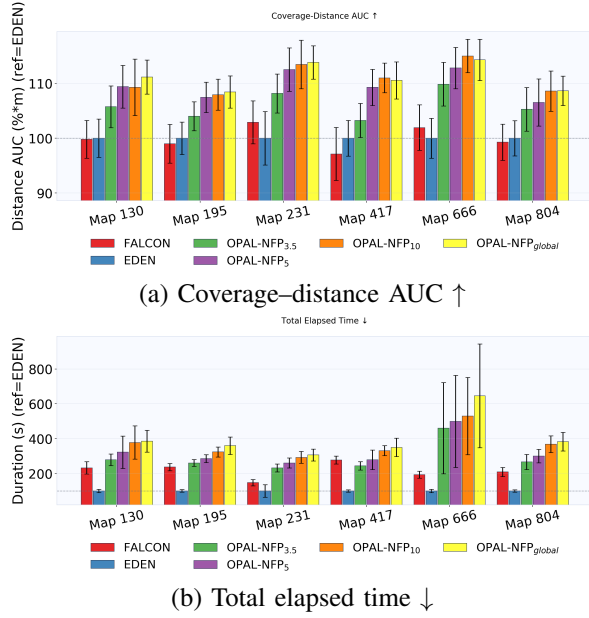


Fig. 3. Reachable-vicinity radius sweep over six maps. (a) AUC is normalized to EDEN, and gains largely plateau near $r_v = 10$ m. (b) EDEN is fastest overall. Within the OPAL-NFP family, elapsed time increases with radius as larger candidate sets trigger more pans. Whiskers show standard deviations.

For the simulation methods, we estimate computation time T_{compute} as $T_{\text{compute}} = T_{\text{elapsed}} - T_{\text{movement}}$, where T_{elapsed} denotes the total exploration time from beginning to end, and T_{movement} denotes the total time the drone is moving, either by translating or rotating.

A. Simulation

Fig. 3 characterizes the effect of r_v for OPAL-NFP with $r_v \in \{3.5, 5, 10\}$ m and with no radius limit. Larger r_v values improve coverage-distance AUC, with gains largely plateauing at 10 m, but at the expense of increased elapsed time. EDEN is fastest overall on every map. Within the OPAL-NFP family, the 3.5 m setting is consistently the fastest, and elapsed time increases with radius. This is because a larger candidate neighborhood creates more ambiguous branch states and therefore triggers more in-place pans before frontier commitment. Across the six maps, OPAL-NFP₁₀ improves AUC by 10.9%, shortens travel distance by 7.0%, and reduces computation time by 53.8% relative to EDEN (Table I). However, the yaw action is costly in that OPAL requires $3.7\times$ EDEN’s elapsed time.

Fig. 4 separates pan and selector effects for OPAL variants. With the same global geometric rule, OPAL-NFP_{global} outperforms no-pan OPAL-NF_{global} in AUC on every map. At matched $r_v = 3.5$ m, the model-based variants using a VLM or LLM perform comparably to OPAL-NFP_{3.5} in AUC. However, they require longer exploration durations than OPAL-NFP_{3.5} because of LLM or VLM query and processing times. Therefore, for the remainder of the paper, we only focus on the geometric variant of OPAL, namely OPAL-NFP.

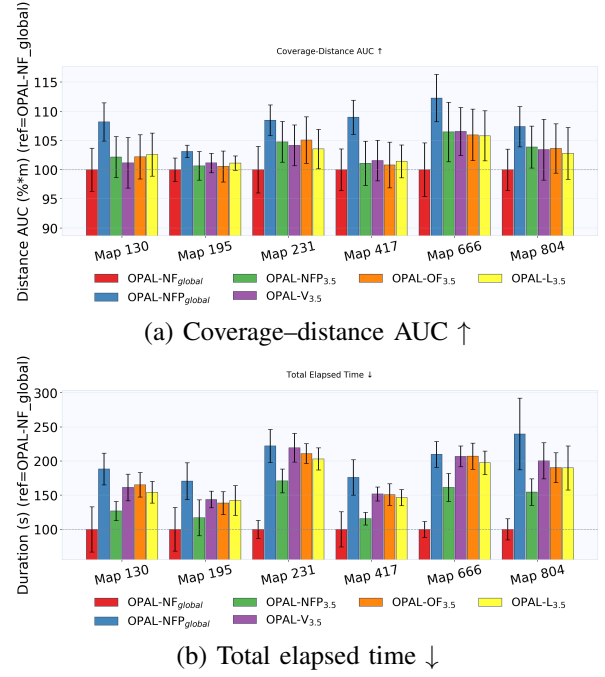


Fig. 4. Pan and selector comparisons over six maps, normalized to no-pan OPAL-NF_{global}. At matched $r_v = 3.5$ m, model-based variants perform comparably to geometric OPAL-NFP_{3.5} in AUC but take longer. The global pan policy incurs the greatest pan time overhead. Whiskers show standard deviations.

1) *Ablation of post-pan updates:* We wish to determine whether the superior distance performance of our proposed approach with pan yaw is due to better depth map building or due to having more frontiers from which to choose. To this end, we perform ablation studies that execute a pan but vary post-pan depth map integration, denoted by M , and post-pan frontier refresh, denoted by F . Let $M1$ denote a map update after a pan and $M0$ denote no post-pan map update. Similarly, let $F1$ denote a post-pan frontier refresh and $F0$ denote no post-pan frontier refresh. Using this notation, $M1F0$ updates only the map, $M0F0$ updates neither the map nor the frontier set, and $M0F1$ refreshes only the frontiers. Fig. 5 shows coverage-distance AUC for $M1F0$, $M0F0$, and $M0F1$ normalized to the OPAL-NFP₁₀ reference. As seen, performance is highest for OPAL-NFP₁₀, followed by the map-only update $M1F0$, no updates $M0F0$, and the frontier-only update $M0F1$. We conclude depth map integration retains more of the reference policy’s distance efficiency than frontier refresh alone, however overlapping standard deviations preclude a significance claim.

B. Hardware experiments

We deploy OPAL on a ModalAI Starling 2 platform with an onboard RGB camera, time-of-flight (ToF) depth sensor, and visual-inertial odometry in two indoor layouts, referred to as Layout 1 and Layout 2. Fig. 6 shows a representative forward RGB observation captured by the onboard camera during an indoor run. Hardware coverage is measured within predefined exploration bounds of $6 \text{ m} \times 9 \text{ m} \times 3 \text{ m}$ for Layout 1 and $5 \text{ m} \times 10 \text{ m} \times 3 \text{ m}$ for Layout 2. As seen in

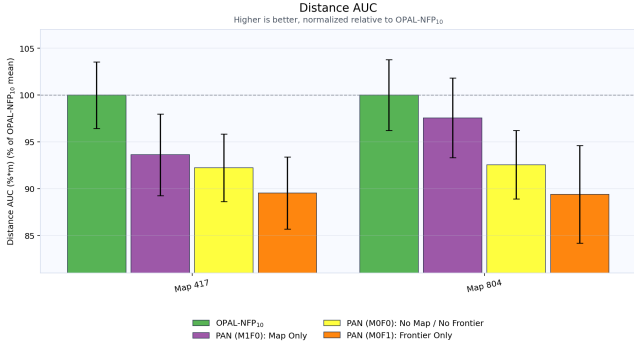


Fig. 5. Coverage-distance AUC ($\uparrow$) for post-pan update conditions on Maps 417 and 804, normalized to the mean of full OPAL-NFP₁₀. Bars show means and whiskers show standard deviations across runs.

TABLE I

COMPARISON OF OPAL-NFP₁₀ WITH EDEN AVERAGED OVER 50 RUNS PER MAP ACROSS SIX MAPS. DELTA IS THE MEAN OF PER-MAP CHANGES.

Metric	EDEN	OPAL	Delta
AUC (% m) $\uparrow$	21026.9	23182.4	+10.9%
Distance (m) $\downarrow$	179.3	167.2	-7.0%
Computation (s) $\downarrow$	53.4	22.7	-53.8%
Elapsed (s) $\downarrow$	145.7	528.1	$3.7\times$

Table II, OPAL achieves higher AUC and shorter travel than FALCON in both layouts, but requires more elapsed time. Furthermore, as expected, OPAL-NFP₁₀ achieves better AUC performance but requires more elapsed time than OPAL-NFP_{3.5}. For Layout 2, the path visualizations in Fig. 7 agree with the travel-distance metric in Table II. OPAL-NFP₁₀ has the shortest path, followed by OPAL-NFP_{3.5} and then FALCON. These traces demonstrate onboard feasibility and reproduce the distance-time trade-off. However, the available records do not include flight counts or uncertainty. The hardware results are therefore illustrative and do not establish statistical reliability or broad real-world superiority.

TABLE II

HARDWARE-VALIDATION RESULTS. Δ IS RELATIVE TO FALCON.

Layout	Method	AUC (% m) $\uparrow$	Distance (m) $\downarrow$	Elapsed (s) $\downarrow$
1	FALCON	2405.6	26.0	36.7
	OPAL-NFP _{3.5}	2647.4	21.9	56.0
	Δ vs. FALCON	+10.1%	-15.8%	+52.6%
2	FALCON	2755.8	32.5	32.0
	OPAL-NFP _{3.5}	2754.6	27.8	41.3
	OPAL-NFP ₁₀	2970.1	24.2	54.0
	Δ OPAL-NFP ₁₀ vs. FALCON	+7.8%	-25.5%	+68.8%

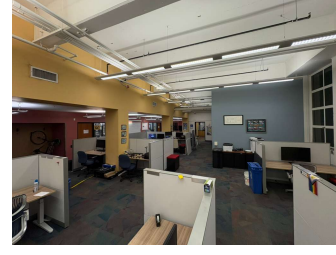


Fig. 6. Representative forward RGB camera observation during an indoor hardware-validation run.

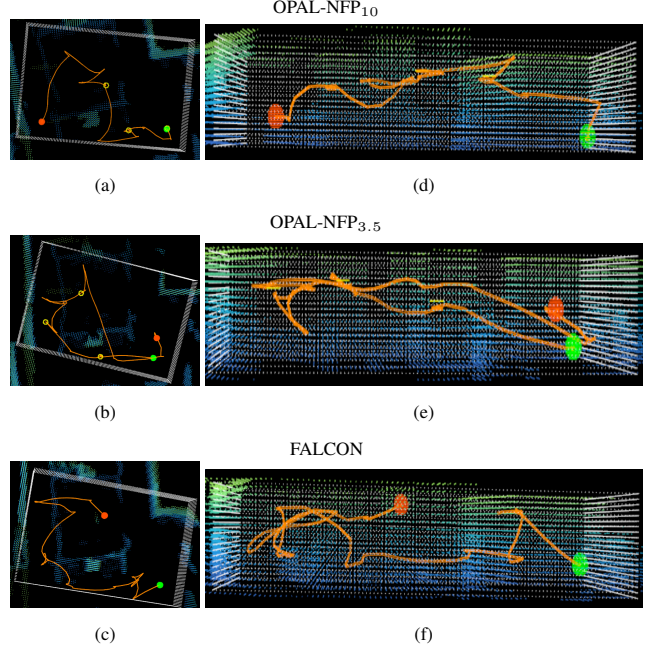


Fig. 7. Representative voxel maps from hardware runs in Indoor Layout 2. Panels (a)–(c) are top-down views of OPAL-NFP₁₀, OPAL-NFP_{3.5}, and FALCON. Panels (d)–(f) are their corresponding side views. The side-view axes use different scales to aid trajectory visualization. Voxel height follows a blue-to-green gradient, with blue closer to the ground. Green and red circles mark the start and end points, the orange trace shows the executed trajectory, yellow discs mark locations where the drone performs an in-place 360° pan, and the white rectangle shows the exploration bounds. The panels are representative rather than matched statistical trials.

IV. CONCLUSIONS

We introduce OPAL, a frontier-exploration framework that adds a 360° yaw pan at ambiguous branch points before selecting the next frontier. Based on simulation results on six maps, OPAL-NFP₁₀ outperforms EDEN in coverage-distance AUC, total traversed distance, and computation time but at the expense of longer elapsed time. The hardware experiments further support this distance-time trade-off. OPAL reduces travel distance relative to FALCON in both layouts but takes longer to complete. One avenue for future work is to mount the depth camera on a motorized rotation mechanism rather than rotating the entire vehicle. This approach could reduce total exploration time and may cost less than a LiDAR-based alternative.

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